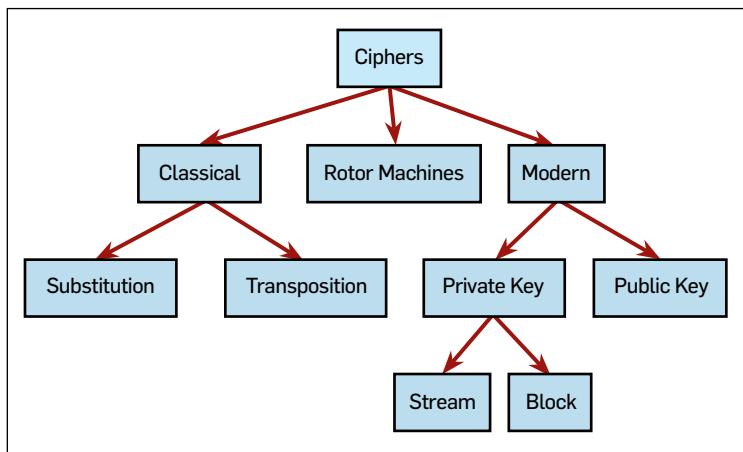




Different kinds of ciphers out there

E-commerce. Now is the time we get down to what exactly prompted us to speculate of such unpredictable times.

So, at the last page, we left the issue of computers at ‘computers are awesome’. Well, they are. But they’re not as awesome as they can be. Not yet.

Let us treat you with an example here.

We give you a number. Say, 287. Now, we tell you to factorize it. Those of you who do not get chills solving class 5th problem at this point (need not worry if you are in class 5th right now, though), might come up with something like $2^{10} + 3^3$. Good on you, sir. What you did was absolutely incredible, if we’re talking in terms of computing power. This is called the factorization problem, and this (though not in such an easy form, obviously) is what is used in all forms of cryptography known to you. Because, as it turns out, our computers cannot factorize that well. In the grand scale of things, they cannot do it at all. And that’s why RSA loves this problem - so much so that their entire work model is structured around this problem. That is the reason the world fears Quantum Computing as much as it lusts for it. The moment someone announces that a full fledged quantum computer has been built, panic would ensue across everything that is based on money and the internet. So basically the global economy.

Why, you ask? Well, as it turns out, a computer that can use qubits (quantum bits) instead of bits can crack this problem in about as much time as it took you to read the last word of this sentence. Interested (and really